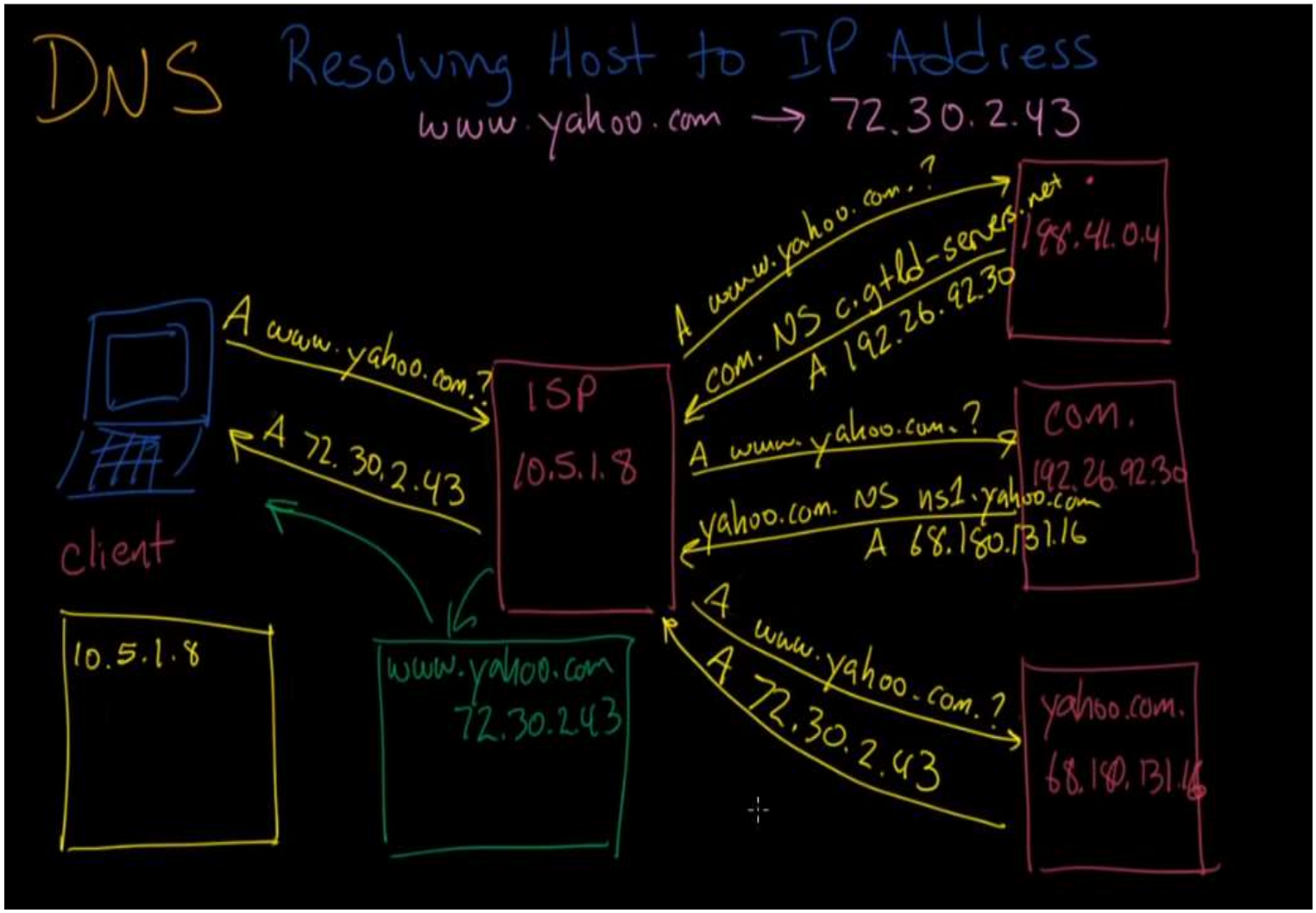
SMTP, POP3, IMAP


DNS
Resolving Host to IP Address
www.yahoo.com → 72.30.2.43
A www.yahoo.com.?
A 72.30.2.43
client
ISP
10.5.1.8
A www.yahoo.com.?
com. NS c.gtld-servers.net
A 192.26.92.30
198.41.0.4
A www.yahoo.com.?
com.
192.26.92.30
yahoo.com. NS ns1.yahoo.com
A 68.180.131.16
A www.yahoo.com.?
A 72.30.2.43
yahoo.com.
68.180.131.16
10.5.1.8
www.yahoo.com
72.30.243

DNS Terminology

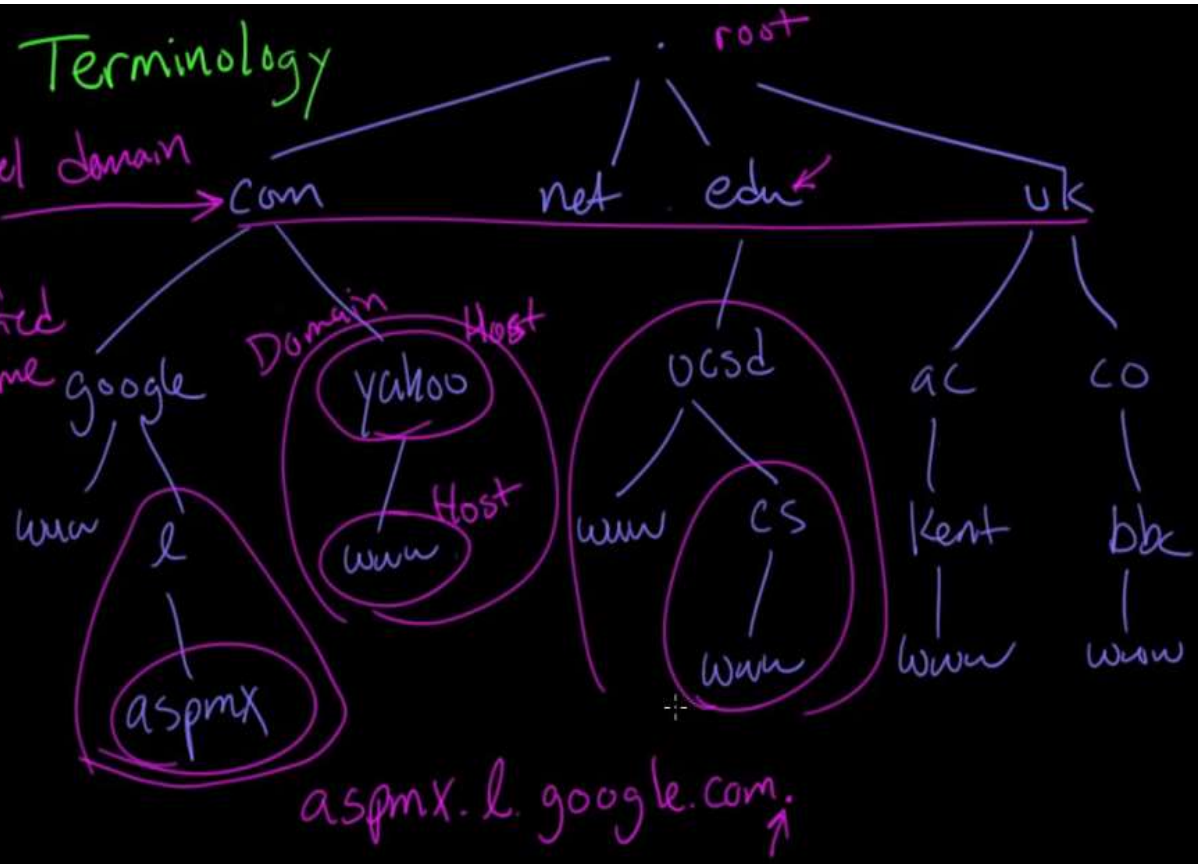
TLD = top level domain

FQDN = Fully qualified domain name

Host

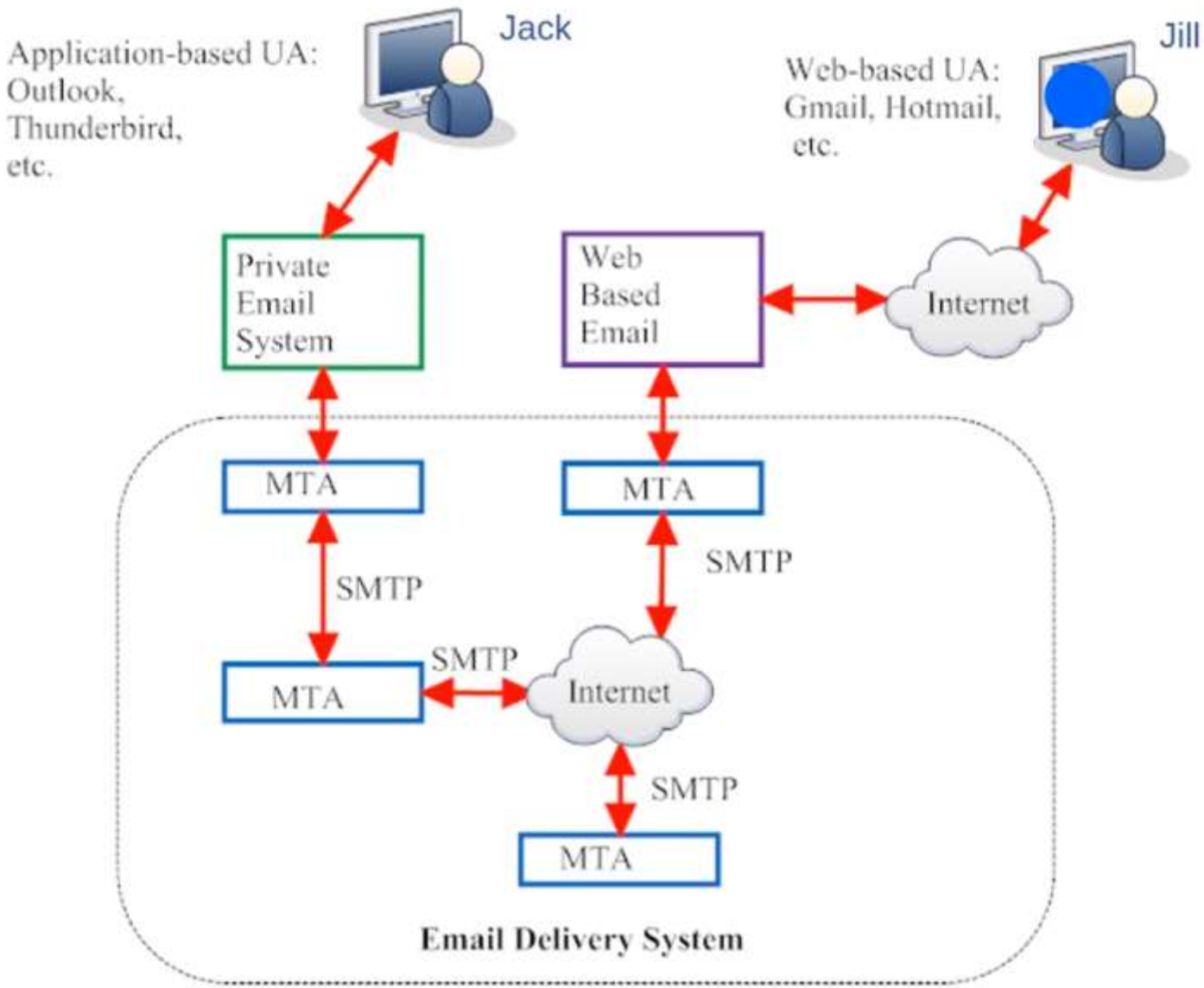
Domain

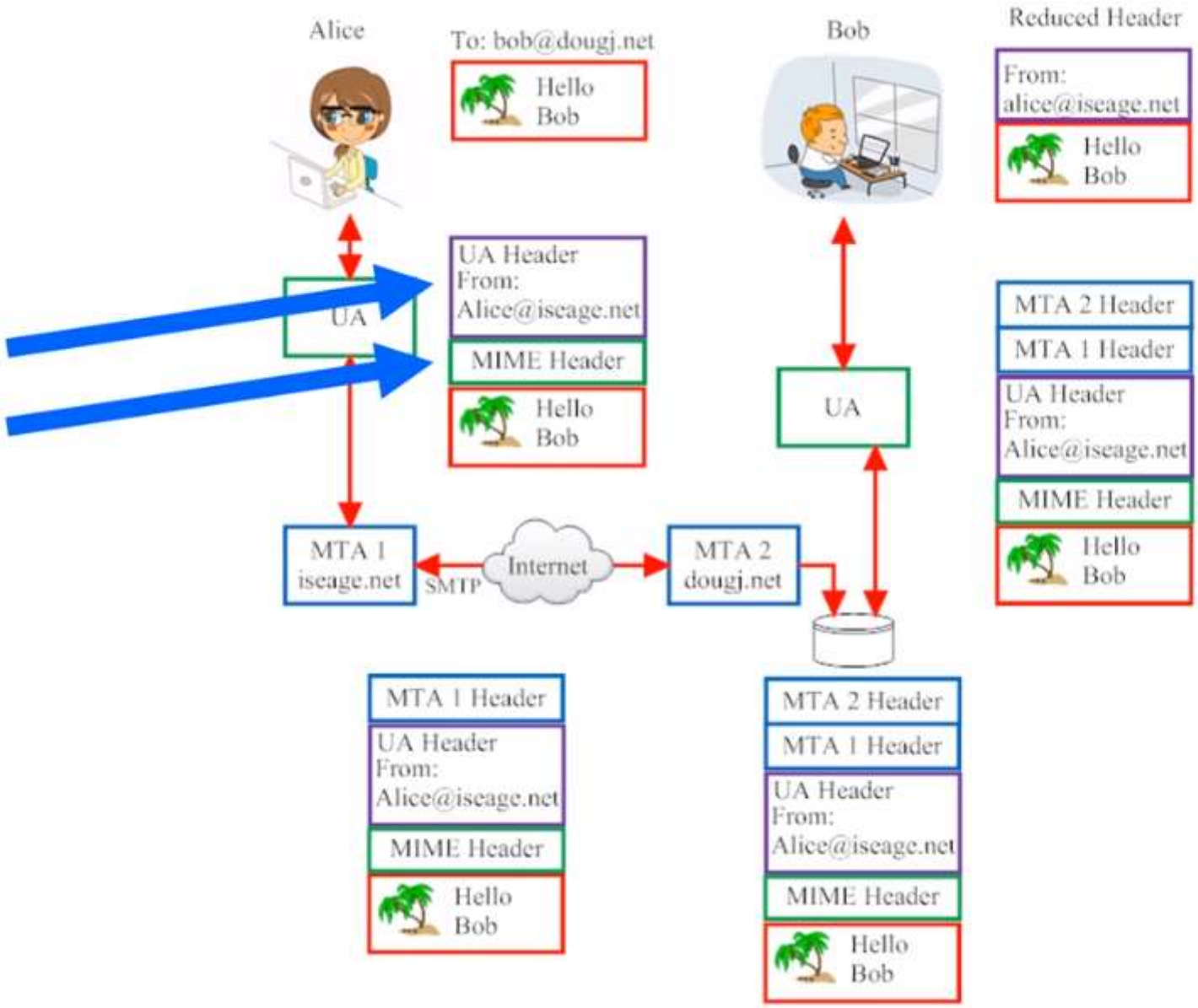
Subdomain



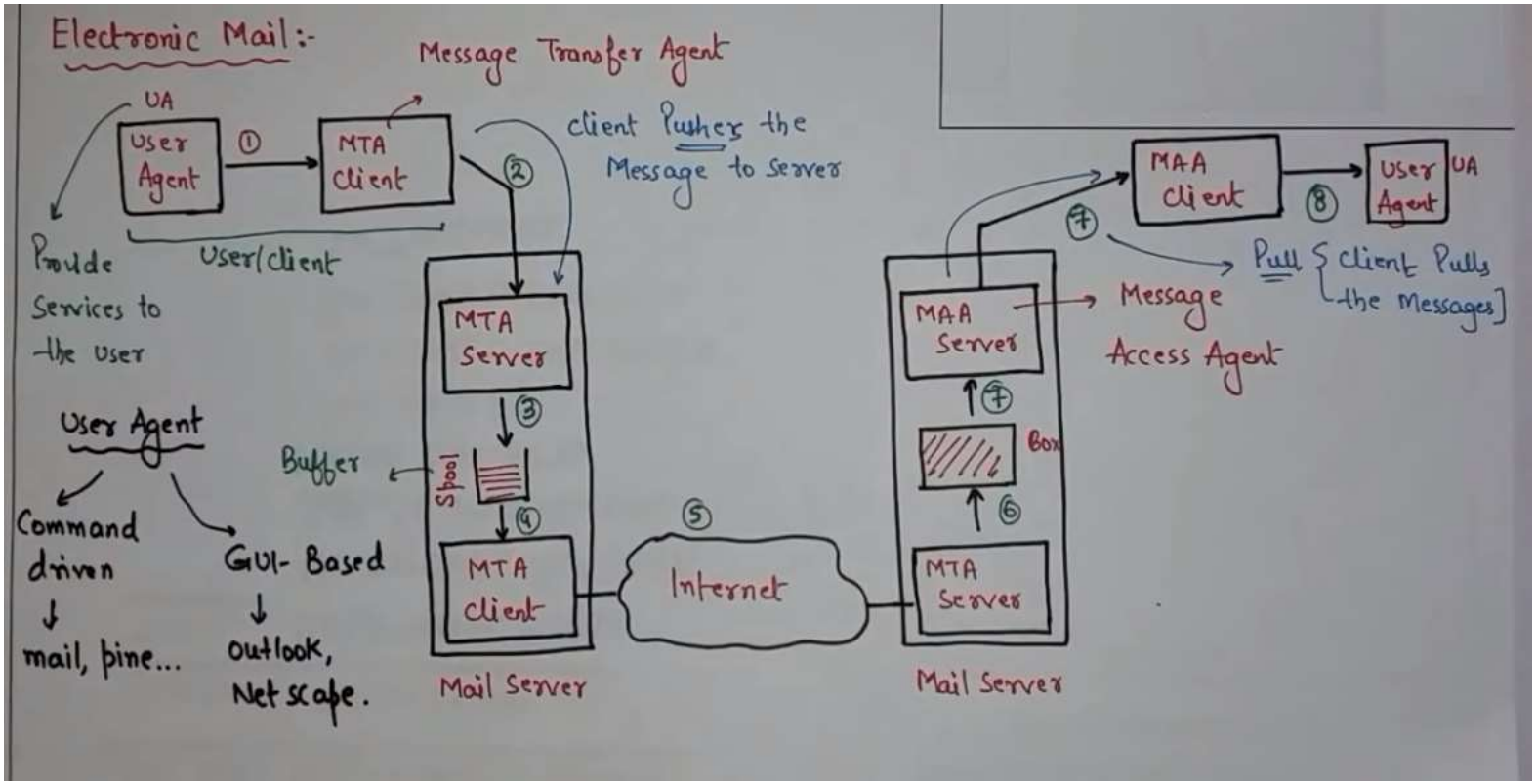
aspmx.l.google.com.

How Email Works




Alice
To: bob@dougj.net
Hello
Bob
Bob
Reduced Header
From:
alice@iseage.net
Hello
Bob
UA
UA Header
From:
Alice@iseage.net
MIME Header
Hello
Bob
UA
MTA 2 Header
MTA 1 Header
UA Header
From:
Alice@iseage.net
MIME Header
Hello
Bob
MTA 1
iseage.net
SMTP
Internet
MTA 2
dougj.net
MTA 1 Header
UA Header
From:
Alice@iseage.net
MIME Header
Hello
Bob
MTA 2 Header
MTA 1 Header
UA Header
From:
Alice@iseage.net
MIME Header
Hello
Bob

Electronic Mail:-



Sending Mail:

Mail From: abc @ xyz.com RCPT TO: efg @ xyz.com	Envelope = Sender and Reciver Address.
From: ABC Corp. To: EFG Corp Date: 15/06/2017 Subject: New Proposal	Header → defines sender, rec, Date, Subject Message
Dear EFG Corp. Thanks ABC	Body = Actual Info^n.

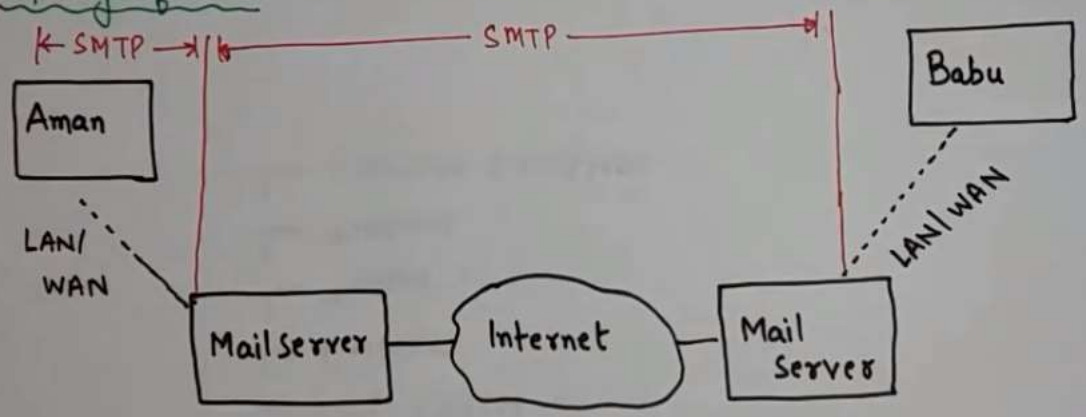
Receiving Mail:

The user agent is triggered by the user (or timer). If user has a mail, the UA informs user with notice.

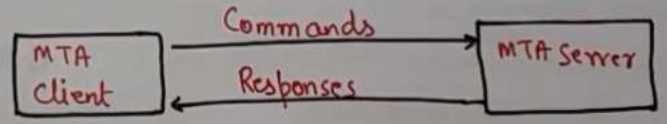
↳ List with Message Subject and Summary is then displayed to the user.

Simple Mail Transfer Protocol (SMTP): It is the protocol that defines MTA client and server in the internet.

Use/Range of SMTP:-



SMTP uses Commands and Responses to transfer Message b/w an MTA client and an MTA server.



Example of Commands:-

- i) HELO → used by client to identify itself.
- ii) MAIL FROM :- identify sender.
- iii) RCPT TO : identify intended receipient
- iv) DATA :- send actual message.
- v) QUIT :- QUIT → Terminate the message
- vi) RSET :- RSET → Reset the Connection
- vii) VRFY :- verify the add. of receipient
- viii) HELP :- HELP: mail

DATA
This is to remind
:
.
VRFY: xyz@efg.com

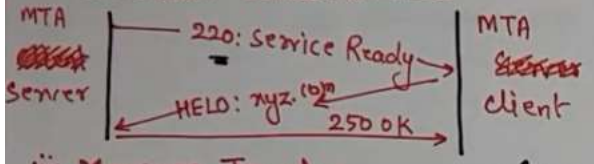
HELO: [redacted] xyz.com → client
MAIL FROM: abc @ xyz.com → sender
RCPT TO: xyz@efg.com

Example of Responses:-

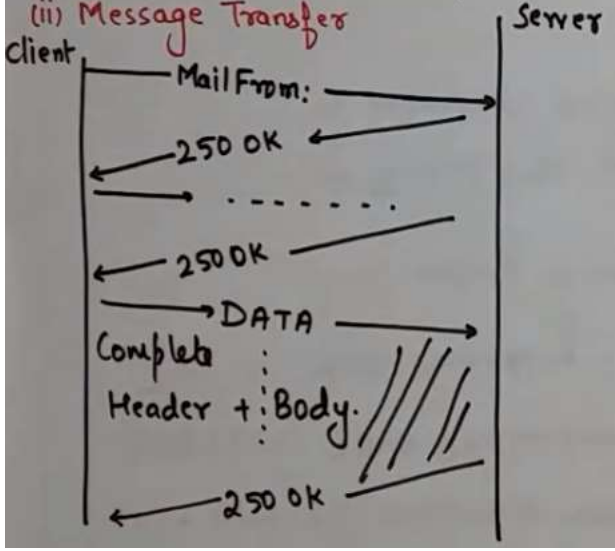
- i) Positive Condition Reply
  - 211 - System Status
  - 214 - HELP
  - 220 - Service Ready
  - 221 - Service Closing
- ii) Transient Negative Completion Reply
  - 421 - Service NA
  - 450 - Mailbox NA
  - 451 - Command Aborted
- iii) Permanent Negative Completion Reply
  - 500 - Syntax Error
  - 554 - Transaction Failed

MAIL TRANSFER PHASES:-

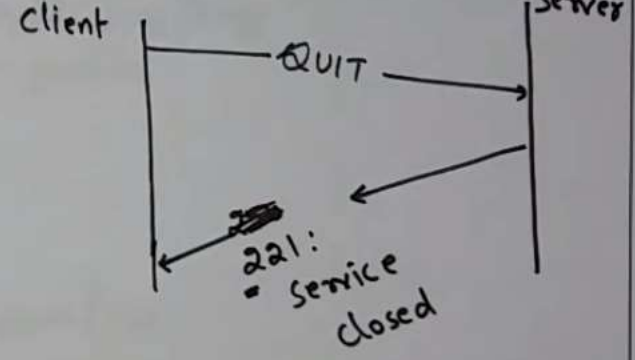
(i) Connection Establishment



(ii) Message Transfer

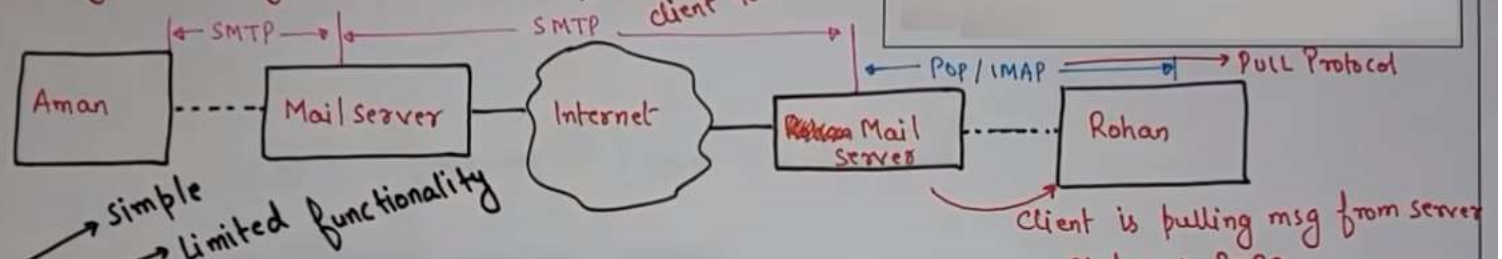


(iii) Connection Termination



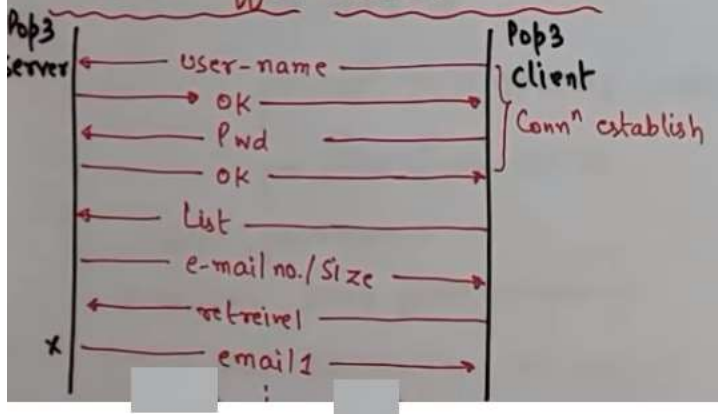
Data Communication and Networking- Application Layer-13

Message Access Agent: POP and IMAP



→ Simple
→ Limited functionality

POP3: Post office Protocol version 3



Modes of POP3:-

Delete Mode

Mail is deleted from the Mailbox after Retreival.

Keep Mode

used when user access mail away from their Comp. Mail is Read but also Kept in the System.

IMAP4: Internet Mail Access Protocol version 4.

IMAP4 is similar to POP3 but with more features. It is more powerful and complex. [IMAPv4 and POP3]. V.IMP.

Extra Features of IMAP4:-

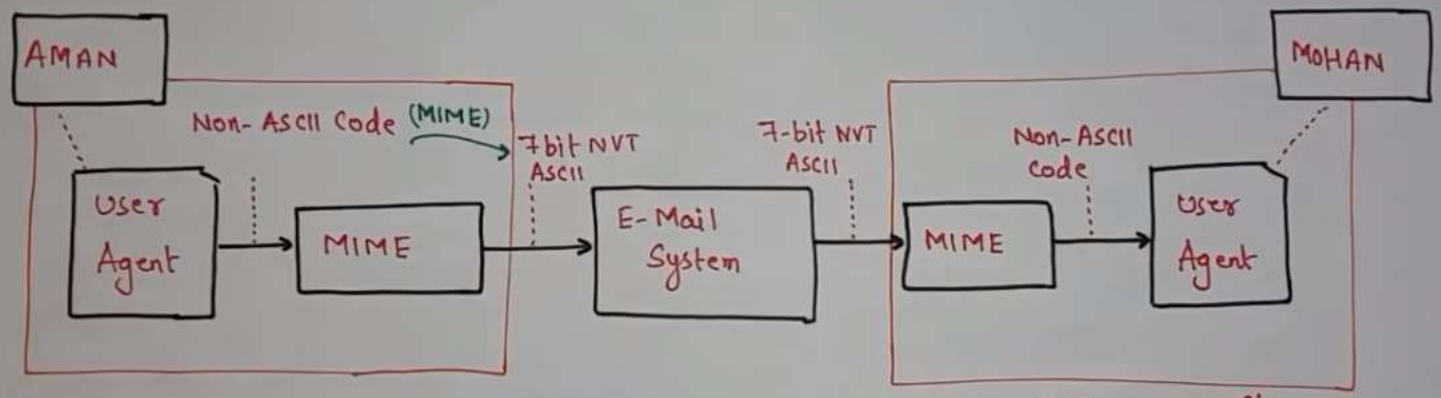
- i) User can partially download email.
- ii) User can create, delete or Rename the mailbox on server.
- iii) User can search contents of the email.
- iv) User can also check e-mail header prior to downloding.
- v) User can create folders to organize the mails in hierarchy.

Disadvantages of POP3:-

- i) Doesnot allow users to organize mails on server.
- ii) No separate folders on server.
- iii) No Partial checking content of mail before downloading.

MIME - Multipurpose Internet Mail Extensions

It is a supplementary Protocol that allows non-ASCII data to be sent through e-mail. (NVT 7-bit ASCII format)



MIME Header:- It is the added to the original e-mail header section to define transformation Parameters:-

- MIME version (1.1)
- Content-Type → Text (Plain, HTML), Video, Audio
- Content-Transfer Encoding → method used for encoding.
- Content-Id → unique identifier
- Content-description → Body is image, video...

POP and IMAP = **both** for inbound email

Application Layer

Post Office Protocol & Internet Message Access Protocol

- POP does not keep the server and client in sync. When you download your mail, it is deleted from the server, so the server is not further updated.
- IMAP keeps the two synced – you only download a **copy**. It is only deleted from the server when you manually delete it.